

Final Project: Database Design and Implementation



Estimated time needed: 90 minutes

Congratulations! You have finished the modules. Now is the time to put your skills to the test. Read through the scenario below.

Scenario

In this scenario, you have recently been hired as a Data Engineer by a New York-based coffee shop chain looking to expand nationally by opening several franchise locations. They want to streamline operations and revamp their data infrastructure as part of their expansion process.

Your job is to design their relational database systems for improved operational efficiencies and make it easier for their executives to make data-driven decisions.

Currently, their data resides in several systems: accounting software, supplier databases, point of sales (POS) systems, and even spreadsheets. You will review the data in all of these systems and design a central database to house all of the data. You will then create the database objects and load them with source data. Finally, you will create subsets of data your business partners require, export them, and load them into staging databases using several RDBMS.

Software used in this project

In this project, you will use [PostgreSQL Database](#), [MySQL](#). These are all relational database management systems (RDBMS) designed to store, manipulate, and retrieve the data efficiently.



Data used in this project

In this project, you will be working with a subset of data from the [Coffee shop sample data](#).

You will use a modified version of the data for the project, so to succeed in the project, download the linked files when prompted in the instructions. You do not need to use any data from the source.

In your scenario, you will be working with data from the following sources:

- Staff information held in a spreadsheet at headquarters (HQ)
- Sales outlet information held in a spreadsheet at HQ
- Sales data output as a CSV file from the POS system in the sales outlets
- Customer data output as a CSV file from a custom customer relationship management system
- Product information maintained in a spreadsheet exported from your supplier's database

Objectives

After completing this lab, you will be able to:

- Identify entities
- Identity attributes
- Create an entity relationship diagram (ERD) using the pgAdmin ERD tool
- Normalize tables
- Define keys and relationships
- Create database objects by generating and running the SQL script from the ERD tool
- Create a view and export the data
- Create a materialized view and export the data
- Import data into a MySQL database using phpMyAdmin GUI tool

Task 1: Identify entities

The first step when designing a new database is to review any existing data and identify the entities for your new system.

1. The following image shows sample data from each source you will be working with to design your new central database. Review the image and identify the entities you plan to create.

staff					
staff_id	first_name	last_name	position	start_date	location
1	Sue	Tindale	CFO	08/03/2001	HQ
2	Ian	Tindale	CEO	3/8/2001	HQ
3	Marny	Hermione	Roaster	10/24/2007	WH
4	Chelsea	Claudia	Roaster	3/7/2003	WH
5	Alec	Isadora	Roaster	2/4/2008	WH
6	Xena	Rahim	Store Manager	7/24/2016	3
7	Kelsey	Cameron	Coffee Wrangler	10/18/2003	3
8	Hamilton	Emi	Coffee Wrangler	9/2/2005	3
9	Caldwell	Veda	Coffee Wrangler	9/9/2013	3
10	Ima	Winifred	Coffee Wrangler	10/12/2016	3

sales_outlet						
sales_outlet_id	sales_outlet_type	address	city	telephone	postal_code	manager
2	warehouse	164-14 Jamaica Ave	Jamaica	972-871-0402	11432	
3	retail	32-20 Broadway	Long Island City	777-718-3190	11106	6
4	retail	604 Union Street	Brooklyn	619-347-5193	11215	11
5	retail	100 Church Street	New York	343-212-5151	10007	16

sales_transaction								
transaction_id	transaction_date	transaction_time	sales_outlet_id	staff_id	customer_id	product_id	quantity	price
1	27/04/2019	09:53:55	8	42	0	38	2	3.75
1	27/04/2019	09:53:55	8	42	0	84	1	0.8
2	27/04/2019	08:00:34	8	42	0	51	2	3
3	27/04/2019	09:04:58	8	42	0	33	1	3.5
4	27/04/2019	08:48:32	8	42	8232	27	1	3.5
5	27/04/2019	09:21:40	8	45	8223	24	1	3

customer						
customer_id	customer_name	customer_email	customer_since	customer_card_number	birthdate	gender
3001	Kelly Key	Venus@adipiscing.edu	04/01/2017	908-424-2890	29/05/1950	M
3002	Clark Schroeder	Nora@fames.gov	07/01/2017	032-732-6308	30/07/1950	M
3003	Elvis Cardenas	Brianna@tellus.edu	10/01/2017	459-375-9187	30/09/1950	M
3004	Rafael Estes	Ina@non.gov	13/01/2017	576-640-9226	01/12/1950	M
3005	Colin Lynn	Dale@Integer.com	15/01/2017	344-674-6569	01/02/1951	M

product					
product_id	product_category	product_type	product_name	description	price
1	Coffee beans	Organic Beans	Brazilian - Organic	It's like Carnival in a cup. Clean and smooth.	18
2	Coffee beans	House blend Beans	Our Old Time Diner Blend	Our packed blend of beans that is reminiscent of the cup of coffee you used to get at a diner.	18
3	Coffee beans	Espresso Beans	Espresso Roast	Our house blend for a good espresso shot.	14.75
4	Coffee beans	Espresso Beans	Primo Espresso Roast	Our premium single source of hand roasted beans.	20.45
5	Coffee beans	Gourmet Beans	Columbian Medium Roast	A smooth cup of coffee any time of day.	15
6	Coffee beans	Gourmet Beans	Ethiopia	From the home of coffee.	21

- Note: You can download a copy of this image or open it in another browser tab for reference later in the lab.

2. Make a list of the entities you have identified. Take a screenshot and save it as Task1.jpg or Task1.png.

Task 2: Identify attributes

In this task, you will identify the attributes of one of the entities you plan to create.

1. Using the information from the sample data in the image from Task 1, identify the entity's attributes that will store the sales transaction data.
2. Make a list of the sales transaction attributes that you identified. Take a screenshot and save it as Task2.jpg or Task2.png.

Task 3: Create an ERD

Now that you have defined some of your attributes and entities, you can determine the tables and columns for them and create an entity-relationship diagram (ERD).

1. Open a new terminal from the side-by-side Cloud IDE.
2. Use the button below to start a PostgreSQL service session in the Cloud IDE.

Open and Start PostgreSQL in IDE

3. Use the pgAdmin weblink to open pgAdmin in a new tab in your browser.
4. Create a new database named COFFEE, view the schemas in the new COFFEE database, and then start a new ERD project.
5. Add a table to the ERD for the sale transactions entity using the information in the following table. Consider the naming convention to use so that your colleagues can understand your data and ensure that the names are valid in other RDBMS. Use the sample data shown in the image in Task 1 to determine appropriate data types for each column.

Sales transaction
Transaction id
Date
Time
Sales outlet
Staff
Customer
Product
Quantity
Price

- 6. Take a screenshot of your ERD and save it as Task3A.png or Task3A.jpg.
- 7. Add a table to the ERD for the product entity using the information in the following table.

Product
Product id
Category
Type
Product
Description
Price

- 8. Take a screenshot of your ERD and save it as Task3B.png or Task3B.jpg.

Task 4: Normalize tables

When reviewing your ERD, you notice it does not conform to the second normal form. In this task, you will normalize some of the tables within the database.

- 1. Review the data in the sales transaction table. Note that the transaction id column does not contain unique values because some transactions include multiple products.
- 2. Determine which columns should be stored in a separate table to remove the repeating rows and to put this table into second normal form.
- 3. Add a new table named `sales_detail` to the ERD, define the columns in the new table, and delete the moved columns from the sales transaction table, leaving a matching column in each of the two tables to create a relationship between them later.
- 4. Take a screenshot of your ERD and save it as Task4A.png or Task4A.jpg.
- 5. Review the data in the product table. Note that the product category and product type columns contain redundant data.
- 6. Determine which columns should be stored in a separate table to reduce redundant data and to put this table into a second normal form.
- 7. Add a new table named `product_type` to the ERD, define the columns in the new table, and delete the moved columns from the product table, leaving a matching column in each of the two tables to create a relationship between them later.
- 8. Take a screenshot of your ERD and save it as Task4B.png or Task4B.jpg.

Task 5: Define keys and relationships

After normalizing your tables, you can define their primary keys and relationships between the tables in your ERD.

- 1. Identify an appropriate column in each table to be a primary key and create the primary keys in the tables in your entity-relationship diagram (ERD).
- 2. Take a screenshot of your ERD and save it as Task5A.png or Task5A.jpg.
- 3. Identify the relationships between the following pairs of tables and then create the relationships in your ERD:
 - `sales_detail` to `sales_transaction`
 - `sales_detail` to `product`
 - `product` to `product_type`
- 4. Take a screenshot of your ERD and save it as Task5B.png or Task5B.jpg.

Task 6: Create database objects by generating and running the SQL script from the ERD tool

Now that your design is complete, you will generate an SQL script from your ERD, which you can use to create your database schema. For this project, you will then use a given SQL script to ensure that you can load the sample data into the schema. Finally, you will load the existing data from various sources into your new database schema.

- 1. Use the Generate SQL functionality in the ERD tool to create an SQL script from your ERD.
- 2. Download the following `GeneratedScript.sql` file to your local computer.
 - [GeneratedScript.sql](#)
- 3. In pgAdmin, open the query tool, upload and open the `GeneratedScript.sql` file from your local computer, and then run the script to create the tables defined in the ERD. Verify that the tables exist in the COFFEE database’s public schema now.
- 4. Take a screenshot of the tables shown in the tree-view pane on the left side of the page and save it as Task6A.png or Task6A.jpg.
- 5. Download the following `CoffeeData.sql` file to your local computer.

- [CoffeeData.sql](#)

6. In pgAdmin, open another instance of the Query tool, upload and open the `CoffeeData.sql` file from your local computer, and then run the script to populate the tables you just created.
7. In pgAdmin, view the first 100 rows of the `sales_detail` table.
8. Take a screenshot of the Data Output pane and save it as `Task6B.png` or `Task6B.jpg`.

Task 7: Create a view and export the data

The external payroll company has requested a list of employees and the locations at which they work. This list should not include the CEO or CFO who owns the company. In this task, you will create a view in your PostgreSQL database that returns this information and export the results to a CSV file.

1. In your COFFEE database, create a new view named `staff_locations_view` using the following SQL:

```
SELECT staff.staff_id,
staff.first_name,
staff.last_name,
staff.location
FROM staff
WHERE "position" NOT IN ('CEO', 'CFO');
```

2. View all the rows returned from the view.
3. Save the query results to a file named `staff_locations_view.csv` on your local computer.
4. Take a screenshot of the view shown in the tree-view pane on the left side of the page with the results in the Data Output pane, and save it as `Task7.png` or `Task7.jpg`.

Task 8: Create a materialized view and export the data

A marketing consultant requires access to your product data in their MySQL database for a marketing campaign. You will create a materialized view in your PostgreSQL database that returns this information and export the results to a CSV file.

1. In your COFFEE database, create a new materialized view named `product_info_m-view` using the following SQL:

```
SELECT product.product_name, product.description, product_type.product_category
FROM product
JOIN product_type
ON product.product_type_id = product_type.product_type_id;
```

2. Refresh the materialized view with data.
3. View all the rows returned from the view.
4. Save the query results to a file named `product_info_m-view.csv` on your local computer.
5. Take a screenshot of the view shown in the tree-view pane on the left side of the page with the results in the Data Output pane, and save it as `Task8.png` or `Task8.jpg`.

Task 9: Import staff_location data into a MySQL database

The external payroll company has asked you to upload the staff location information to their MySQL database.

1. Click the button below to start a MySQL instance in the Cloud IDE.

Open and Start MySQL in IDE

2. Open phpMyAdmin in a new tab in your browser.
3. In phpMyAdmin, create a new database named `STAFF_LOCATIONS`, then import the location information saved in the `staff_locations_view.csv` file you exported from the view you created in Task 7.
4. Explore the new table and then view the data in it.
5. Take a screenshot of the contents of the new table and save it as `Task9.png` or `Task9.jpg`.

Task 10: Import coffee_shop_products data into a MySQL database

The marketing consultant has asked you to upload the product information to their MySQL database.

1. In phpMyAdmin, create a new database named `coffee_shop_products`, and then import the product information saved in the `product_info_m-view.csv` file from your materialized view into a new table in the `coffee_shop_products` database.
2. Browse the contents of the new table.

3. Take a screenshot of the contents of the new table and save it as Task10.png or Task10.jpg.

Task 11[Optional]: Import staff_location data into Db2 database

The external payroll company has asked you to upload the staff location information to their Db2 database.

1. In a new browser tab, go to cloud.ibm.com/login, log in using your credentials, and then open a console for your Db2 on the Cloud instance you created earlier in this course.
2. Use the Load Data feature to load a new table named STAFF_LOCATIONS with the staff location information saved in the staff_locations_view.csv file you exported from the view you created in Task 7.
3. Explore the new table and then view the data in it.
4. Take a screenshot of the contents of the new table and save it as Task9.png or Task9.jpg.

Recap

In this project, you created the objects and views that you exported in a MySQL database.

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